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PI Name:	O'Connor, Kim C. Ph.D.		
Project Title:	Spatial Organization within Prostate Cancer Spheroids		
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Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Biotechnology		
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Comments:			
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
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Key Personnel Changes/Previous PI:	O'Connor, Kim		
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Multicellular spheroids have a remarkable capacity to mimic solid tumors to a greater extent than cell monolayers. Spatial organization within spheroids is seminal to their in vivo-like behavior and is instrumental in the acquisition of drug resistance and expression of antigens that can serve as biomarkers for carcinogenesis. Our research provides a comprehensive analysis of the spatial profile and assembly of multicellular spheroids, using prostate cancer cells as a model system. Spheroid differentiation and viability will be determined as a function of radial position. This analysis will be performed on spheroids produced using conventional methods and in a rotating-wall vessel (RWV). The knowledge gained from this research will ultimately improve the fidelity of multicellular spheroids in the extent to which they resemble micrometastases and avascular regions of larger tumors. In future work, we will use the spheroids to develop biomarkers for carcinogenesis and countermeasure to against the harmful effects of radiation. The objective of the proposed research is to improve our understanding of how culture conditions affect the spatial profile and assembly of multicellular cancer spheroids, using prostate cancer cells as a model system. During year four of funding,

Task Description:	our specific aims were to: 1) Deconvolute the physical mechanisms underlying the spatial organization of cell aggregates into spheroids, a process known as restructuring dynamics. 2) Characterize the spatial profile of viability, differentiation and expression of multidrug resistance proteins within DU 145 and LNCaP spheroids. 3) Investigate the involvement of intercellular bridges in spheroid formation. NASA Critical Path Roadmap The research described in this report is categorized in the NASA Critical Path Roadmap under the discipline of Radiation Health and addresses the risk of carcinogenesis in astronauts during the exploration of space from exposure to harmful radiation or from the synergistic effects of radiation with other space flight factors including stress. The relevant enabling question for this project is 31d: How can effective biomarkers of carcinogenic risk from space radiation be developed and validated? The overall goal of this four-year project is to produce in vitro cultures of multicellular spheroids that mimic solid tumors by gaining an understanding of tissue assembly and spheroid composition. Through this understanding, we will identify culture conditions favorable to spheroid production. In future work, we will use the spheroids to develop biomarkers for carcinogenesis and countermeasure to against the harmful effects of radiation. Background For NASA, spheroids have been the primary form of tissue reconstructed in RWV's. For the larger tissue engineering community, spheroids have a variety of applications from liver regeneration to stem-cell differentiation. When spheroids are prepared from neoplastic cells as in the proposed research, they mimic micrometastases and avascular regions of larger tumors. Spheroids have proven beneficial as a research tool to investigate cancer progression, identify biomarkers of carcinogenesis, and in the testing of new therapies and delivery strategies to combat this deadly disease. Numerous studies have demonstrated that the RWV produces multicellular spheroids that are more differentiated than cultures of the same cell line grown as a monolayer in terms of spheroid morphology and expression of proteins that are indicative of differentiated function. This is attributed to gentle mixing in the RWV that promotes mass transfer of nutrients and wastes while minimizing cellular damage from hydrodynamic forces; three-dimensional (3D) organization of cells that produces an environment in which cell shape, cell-cell and cell-matrix interactions, and interstitial fluid are akin to those in native tissue; and, for co-cultures, co-location of dissimilar cell types, mimicking the diversity found in vivo. Understanding the mechanisms of spheroid assembly will ultimately improve the fidelity of this reconstructed tissue. The kinetic models developed in this work will be useful in producing spheroids within a defined size range that have greater viability. For, neoplastic cells, model results can also aid in the analysis of cellular adhesion and the effectiveness of new therapies to inhibit tumor shedding. A critical component in the resemblance of spheroids to solid tumors is the spatial profiles within spheroids as a result of mass transport limitations and the 3D environment within the tissue. For example, the region of quiescent cells that lies beneath proliferative cells on the spheroid surface is often instrumental in the failure of tumor regression therapies. These cells have both limited accessibility and intrinsic resistance to toxic agents. Despite the importance of the spatial profile of spheroids to their function, there has been very little research done in this area with RWV's. In fact, ours will be by far the most comprehensive study of its kind for RWV spheroids.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Prostate cancer is one of the most common forms of cancer in men and a leading cause of death in the United States. While surgery and radiation will successfully treat many patients with localized disease, there is still no cure for patients with advanced, metastatic prostate cancer. The proposed research will aid in the development of new cancer therapies and biomarkers for detect prostate cancer by producing tissue models that more closely approximate micrometastases than is currently possible with conventional cell-culture techniques.
Task Progress:	Song, David, Clejan, Giordano, Pappas-LeBeau, Xu, O'Connor*. Spatial composition of prostate cancer spheroids in mixed and static cultures. Tissue Eng, 10: 1266-76 (2004). Aggregation of neoplastic cells produces multicellular spheroids resembling micrometastases. The objective of this study is to investigate the effects of mixing culture medium on the spatial composition of spheroids prepared from well (LNCaP) and poorly (DU 145) differentiated human prostate cancer cells. Spheroids were cultured in a mixed suspension within a high aspect rotating-wall vessel and static liquid-overlay plate. Results from this study demonstrate that mixed cultures consistently manifested differences in morphology and composition between DU 145 and LNCaP spheroids. For example, 40% ± 12% of DU 145 cells were Ki-67 positive 100 mm from the surface within mixed spheroids versus 0% for LNCaP cells; there was no significant difference in this spatial profile for static cultures. The results suggest that poorly differentiated spheroids may be more likely to experience a change in composition from mixing culture medium than well-differentiated spheroids due to low tissue density. Immunostaining for P-glycoprotein is representative of this trend; average staining intensity increased 50% for DU 145 spheroids upon mixing but was unchanged for LNCaP spheroids. The effects of mixing on spheroid composition were attributed to faster interstitial mass transport. Applications include drug development and delivery, as well as basic research on drug action and resistance. Vidulescu, Clejan, O'Connor*. Vesicle traffic through intercellular bridges in DU 145 human prostate cancer cells. J Cell Mol Med, 8: 388-96 (2004). Fluorescence microscopy detected cell-to-cell communication via intercellular bridges in DU 145 human prostate cancer cells. Since DU 145 cells have deficient gap junctions, intercellular bridges may have a prominent role in the transfer of chemical signals between these cells. In culture, DU 145 cells physically interact over several cell diameters through filopodial extension and with adjacent cells across intercellular bridges. These structures range from 100 nm to 5 microns in diameter and a few microns to at least 50 to 100 microns in length. Time-lapse imagery revealed that (1) filopodia rapidly extend at a rate of microns per minute to contact neighboring cells and (2) intercellular bridges are conduits for transport of membrane vesicles (1 to 3 microns in diameter) between adjacent cells. Immunofluorescence detected alpha-tubulin in intercellular bridges and filopodia greater than a micron in diameter. The functional meaning, interrelationship of these membrane extensions are discussed, along with the significance of these findings for other culture systems such as stem cells. Applications of this work include the development of anti-cancer therapies that target intercellular communication and formation of cancer spheroids for drug testing.

	Song, Jain, Enmon, O'Connor*. Restructuring dynamics of DU 145 and LNCaP prostate cancer spheroids. In Vitro Cell Dev Biol - Anim, 40 (9), in press (2004). Neoplastic cells acquire multidrug resistance as they assemble into multicellular spheroids. Image analysis and Monte Carlo simulation provided insight into the adhesion and motility events during spheroid restructuring in liquid-overlay culture of DU 145 and LNCaP human prostate cancer cells. Irregularly shaped, two-dimensional aggregates restructured through incremental cell movements into three-dimensional spheroids. Of the two cultures examined, restructuring was more pronounced for DU 145 aggregates. Motile DU 145 cells formed spheroids with a minimum cell overlay of 30% for 25-mers as estimated by simulation vs. 5% for adhesive LNCaP cells in aggregates of the same size. Over 72 h, the texture ratio increased from 0.55 ± 0.05 for DU 145 aggregates with projected areas exceeding 2000 sq. microns to a value approaching 0.75 ± 0.02 ($p < 0.05$). For LNCaP aggregates of comparable size, the increase in texture ratio was more modest, less than 15% during the same time period ($p < 0.05$). Combined these data suggest that motility events govern the overall rate of spheroid restructuring. This information has application to the chemosensitization of solid tumors and kinetic modeling of spheroid production. Song, O'Connor*, Lacks, Enmon, Jain. Monte Carlo simulation of LNCaP human prostate cancer cell aggregation in liquid-overlay culture. Biotechnol Prog, 19: 1742-9 (2003). The research described in this publication was summarized in the 2003 NASA Annual Report. The article was listed in press in the 2003 Report and was published later in the year.
Bibliography Type:	Description: (Last Updated: 04/03/2007)
Abstracts for Journals and Proceedings	Song, H.; O'Connor, K.; Lacks, D.; Enmon, R.; Jain, S. " Monte Carlo simulation of LNCaP human prostate cancer cell aggregation in liquid-overlay culture. " Society for In Vitro Biology, San Francisco, CA In Vitro Cell Dev Biol - Animal, 40: 35-A , Jun-2004
Articles in Peer-reviewed Journals	Song, H.; Jain, S.; Enmon, R.; and O'Connor, K.C. "Restructuring dynamics of DU 145 and LNCaP prostate cancer spheroids." In Vitro Cell Dev Biol - Anim, 40 (9), in press , Oct-2004
Articles in Peer-reviewed Journals	Vidulescu, C.; Clejan, S.; O'Connor , K. C. " Vesicle traffic through intercellular bridges in DU 145 human prostate cancer cells." J Cell Mol Med, 8: 388-96 , Sep-2004
Articles in Peer-reviewed Journals	Song, H.; David, O.; Clejan, S.; Giordano, C.L.; Pappas-LeBeau, H.; Xu L.; O'Connor, K. C. "Spatial composition of prostate cancer spheroids in mixed and static cultures. " Tissue Eng, 10: 1266-76 , Jul-2004
Articles in Peer-reviewed Journals	Song, H.; O'Connor, K. C.; Lacks, D. J.; Enmon, R. M.; Jain, S. K. "Monte Carlo simulation of LNCaP human prostate cancer cell aggregation in liquid-overlay culture." Biotechnol Prog, 19: 1742-9 , May-2004